Ebbinghaus-Curve Guided Low-Rank Component-Induced Attention for Multisource Remote Sensing Classification

Weiying Xie, *Senior Member, IEEE*, Yibing Lu, Daixun Li, and Yunsong Li, *Member, IEEE*

***Abstract*— The integration of multisource remote sensing (RS) data is crucial in land use and land cover (LULC) studies, offering numerous applications. Using diverse data sources enhances the accuracy of land cover classification. However, due to differences in imaging mechanisms, existing methods face challenges in capturing complex local and global relationships. Moreover, current multimodal fusion approaches often fail to efficiently preserve heterogeneous data, leading to the overfusion of redundant features. To address these challenges, we propose Ebbinghaus-curve guided multisource RS classification network (ECNet). This framework maximizes the benefits of convolutional operators for local feature representation and leverages Transformer architecture for learning long-distance dependencies. Inspired by the forgetting strategy of human brain neurons, we extend the concept of information loss to address feature preservation issues in neural networks. ECNet effectively preserves essential features while eliminating redundancy in high-dimensional manifold structures, thus mitigating overfitting caused by redundant multimodal features. Extensive experiments on four publicly available datasets demonstrate the competitiveness of ECNet in classification tasks. Notably, on the Houston2013 dataset, ECNet achieves an impressive overall accuracy (OA) of 96.63%, surpassing various state-of-the-art baseline approaches. The code is available at https://github.com/lyb10087/ECNet for the sake of reproducibility.**

***Index Terms*— Deep learning (DL), Ebbinghaus forgetting curve, feature fusion, multimodal, remote sensing (RS), transformer.**

I. Introduction

IN RECENT years, remote sensing (RS) technology has advanced significantly, leading to the availability of abundant multisource, multimodal data, including hyperspectral imagery (HSI), multispectral imagery (MSI), light detection and ranging (LiDAR), and synthetic aperture radar (SAR). These varied data types present innovative opportunities for Earth observation. Different sensors can capture simultaneous images of the same geographic area [1]. However, due to disparities in spatial resolution, spectral frequency, and target characteristics among the sensors, the data they provide is both redundant and complementary [2]. HSI offers a detailed spectral information about the Earth's surface but is limited by lower spatial resolution. LiDAR, on the other hand, can acquire elevation information of objects, complementing the capabilities of HSI. SAR furnishes geometric information on ground features in terms of amplitude and phase [3]. Combining HSI and SAR can leverage the unique advantages of optical imaging and active sensing, resulting in a more comprehensive RS observation of the Earth's surface. Data fusion highlights the identification of relevant thematic information, thereby augmenting the reliability of image interpretation [4].

With the substantial progress of deep learning (DL) in various applications, deep neural networks (DNNs) have become extensively employed for addressing classification tasks. Among them, convolutional neural networks (CNNs) have emerged as a cornerstone in this area. In recent years, numerous CNNs methods have been proposed to integrate and leverage multisource RS data. Xu et al. [5]

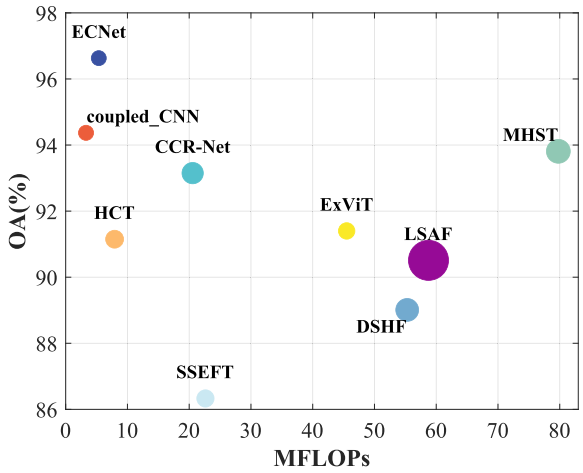


Fig. 1. Comparison of networks in terms of OA, Parameters, and FLOPs on the Houston2013 dataset.

Received 9 August 2024; revised 9 September 2024; accepted 9 October 2024. Date of publication 16 October 2024; date of current version 29 October 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62121001, Grant 62322117, Grant 62371365, and Grant U22B2014; in part by the Young Elite Scientist Sponsorship Program by the China Association for Science and Technology under Grant 2020QNRC001; in part by the Fundamental Research Funds for the Central Universities under Grant QTZX23048; and in part by the Innovation Fund of Xidian University under Grant YJSJ24017. *(Corresponding author: Weiying Xie.)*

The authors are with the State Key Laboratory of Integrated Services Networks, Xidian University, Xi'an 710071, China (e-mail: wyxie@xidian.edu.cn; yibinglu@stu.xidian.edu.cn; ldx@stu.xidian.edu.cn; ysli@mail.xidian.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3482277

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:56:22 UTC from IEEE Xplore. Restrictions apply.

originally introduced a dual-branch CNN network for improving classification performance through multisource integration. Zhang et al. [6] proposed a novel interleaved perceptual CNN to integrate heterogeneous information and improve the accuracy of joint classification. Wu et al. [7] employ a cross-modal reconstruction strategy to compactly integrate and represent RS data from different sensors, facilitating more efficient information exchange. Mei et al. [8] employ CNN for image classification preprocessing, aimed at learning rotation-invariant feature. Hong et al. [9] proposed a unified multimodal DL framework with a focus on RS data classification. They extensively investigated and summarized multistage multimodal fusion strategies, providing a foundational framework for multimodal feature learning.

The aforementioned methods are all based on CNNs and their derivatives. Despite the robust capabilities of CNNs in extracting spatial structural information and local contextual details from multisource RS images, they face limitations in handling HSI data. HSI contains extensive spectral sequence information, posing difficulties in distinguishing subtle spectral differences between pixels. Furthermore, due to the inherent constraints of convolutional operations with finite receptive fields, the necessity for multiple stacked convolutional-pooling structures to expand the focus region adversely impacts the model's ability to effectively capture global information effectively. Transformers [10] demonstrate significant advantages in capturing long-term dependencies and deep global features. In the context of collaborative land cover classification, establishing remote dependencies proves to be an effective means of capturing spectral information and global background related to land cover.

Hong et al. [11] introduced a novel spectral Transformer model designed for processing sequential data, effectively capturing spectral features from adjacent bands. However, this study predominantly concentrates on the spectral information processing of HSI, exhibiting a deficiency in harnessing spatial information. Therefore, Sun et al. [12] proposed the spectrum-space feature tagging transformer. Their objective was to enhance both network weight equilibrium and classification performance. They achieved this by integrating a module for spectral-space feature extraction and a Gaussian-weighted feature tagger. Mei et al. [13] proposed a group-aware hierarchical transformer, which progressively reduces the number of channels in the feature maps through a hierarchical structure, thereby enhancing the semantic information within the feature maps. Due to the distinctive 3-D cuboid structure inherent in HSI data, which encompasses rich spectral and spatial information, Transformer do not align seamlessly with HSI data. Currently, while some Transformer-based approaches address the redundancy in spatial features, most methods overlook redundancy in global sequential features. Furthermore, the linear projection and flattening operations within Transformer can compromise local spatial–spectral and positional information, resulting in the loss of valuable local information during the classification process.

A naturally arising question pertains to how to effectively explore the correlation and complementarity between local spatial and global spectral features in HSI. Concurrently, addressing the issue of feature redundancy is essential to prevent potential performance pitfalls in local optima. The Ebbinghaus forgetting curve delineates the forgetting patterns of human brain neurons. By understanding these forgetting patterns through the temporal function of information decay, self-memory capabilities can be improved. Inspired by the forgetting mechanism in human brain neurons, we apply it to neural networks to address overfitting due to redundant multimodal features. We propose a novel framework: the Ebbinghaus-curve guided CNN-transformer cascaded multimodal framework. To the best of our knowledge, this constitutes the first-ever exploration within the domain of multimodal fusion addressing the issue of redundancy associated with complementary features. Our primary contributions are outlined as follows.

1) We present Ebbinghaus-curve guided multisource RS classification network (ECNet), a sophisticated end-to-end multimodal feature fusion framework that integrates Transformer and CNN through collaborative interaction. ECNet incorporates two crucial components: the spatial information augmentation module (SIAM) and the channel redundancy refinement module (CRRM). These two modules alleviate spectral variations and diminish redundant information by weighting the pixel space and spectral background.
2) A compact collaborative classification model for multisource RS data is designed. The weight-sharing strategy is employed for feature extraction, which reduces computational complexity and improves model execution efficiency (shown in Fig. 1).
3) We propose a manifold feature preservation mechanism guided by the Ebbinghaus forgetting curve. This mechanism achieves a low-rank decomposition of high-dimensional manifold structures while preserving crucial features.
4) Through extensive data validation, our proposed framework has demonstrated significant achievements in handling HSI and LiDAR data. Furthermore, we have successfully extended this framework to more challenging scenarios involving HSI and SAR data, achieving remarkable results.

II. Related Works

A. RS Images Multimodal Fusion

Researchers persistently endeavor to develop effective methods for integrating information gathered from various sensors. Their goal is to enhance the efficiency of processing multimodal data. Within the domain of DL-based multisource data fusion, fusion methodologies are mainly classified into three categories: pixel-level fusion, feature-level fusion, and decision-level fusion. Simple pixel-level fusion techniques [14] often exhibit low efficiency and are susceptible to the Hughes phenomenon. This phenomenon denotes a performance decline resulting from merging features at the raw or pixel level without considering their underlying statistical properties. To address this issue, feature-level fusion methods emerged.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:56:22 UTC from IEEE Xplore. Restrictions apply.

The feature-level fusion method aims to extract and aggregate features from multiple RS data sources, encompassing spatial characteristics such as shape, edges, and textures, as well as spectral features like reflectance curves and peak positions. The fundamental concept involves a fusion algorithm based on regional blocks. Benedetti et al. [15] DL-based generic framework propelled the fusion applications of multispectral and multimodal satellite data in diverse temporal and spatial dimensions. Xi et al. [16] introduced an innovative deep generative spectral-space classifier. This classifier demonstrated outstanding performance when dealing with complex ground scenes.

Feature-level fusion not only effectively integrates information such as spectral, spatial, and elevation, thereby enhancing the data's representational capacity, but also strengthens the system's capability to capture subtle features. Simultaneously, decision-level fusion enhances the overall system's robustness and generalization by effectively combining decision information from different models or methods. This makes the system more reliable, especially in complex scenarios. In our investigation, we exploited the benefits of feature-level and decision-level fusion, seamlessly amalgamating them for the categorization of multimodal RS data, yielding noteworthy research outcomes.

B. Low-Rank Decomposition SL Strategies

Currently, most DL-based methods that achieve high classification accuracy rely heavily on a large amount of labeled data. However, acquiring these labeled samples for HSI data can be both time-consuming and challenging, and in some cases, it may even be infeasible. The high dimensionality of HSI data also poses challenges for the design of classification methods. This is known as the high-dimensional small-sized problem. In the analysis of HSI data, spectral bands often exhibit high interband correlation, which can be removed without losing important information. Therefore, subspace learning (SL) [17] as a preprocessing step is crucial for HSI classification to reduce interband redundancy while preserving important discriminative information for subsequent classification tasks.

Dimensionality reduction as a primary strategy within SL has been widely adopted. PCA, as an unsupervised method, effectively reduces dimensionality but fails to capture the intrinsic nonlinear structure of HSI data. Nonlinear techniques, such as Laplacian eigenmaps (LEs) [18] and local linear embedding (LLE) [19], have been successfully applied to HSI due to their ability to preserve the geometric structure of data manifolds. However, existing methods mostly utilize spectral information from HSI and do not fully exploit effective features in the spatial domain.

Therefore, low-rank decomposition emerges as a potential solution. Lu et al. [20] introduced an enhanced preserving projection-based method which specifically emphasizes low-rank representations in image classification. Peng et al. have successively proposed the joint sparse representation (JSR) method [21] and self-paced JSR (SPJSR) model [22]. These models demonstrate superior performance in significantly reducing the impact of nonuniform neighboring pixels or outliers, particularly in the presence of heavy noise. They provide a baseline approach for low-rank representation of hyperspectral data. Li et al. [23] compress the feature matrix explicitly into a low-rank subspace using cascading and additive approaches. This highlights the manifold structure of shallow features. Yuan et al. [24] proposed an innovative low-rank and half-nonnegative tensor decomposition method for denoising HSI. They introduced framework regularization to constrain the dimension reduction factors, effectively preserving the details and geometric features of the HSI in the spatial domain. Lu et al. [25] proposed a method based on a separable deep graph convolutional network, which enhances feature representation by using prototype learning with a regularization term to improve intraclass compactness and interclass separability. However, the aforementioned methods have failed to achieve adaptive recognition of homogeneous neighborhoods, thus unable to map effective spectral information onto nonlinear and complex manifolds. To solve this problem, we propose the forgetting curve low rank resolver (FCLR). By synergistically employing PCA and SVD, in conjunction with the low-rank matrix recovery (LRMR) technique, we attain meticulous spectral reconstruction for HSI.

III. Methodology

A. Framework of the Proposed Model

Fig. 2 illustrates the overall architecture of the ECNet, comprising four primary components: The spatial–spectral attention mechanism (SSAM) for extracting spatial and spectral features from HSI, the SIAM, and CRRM for multimodal data fusion, and the FLCR module, which leverages the forgetting curve to preserve features and eliminate redundant structures within high-dimensional manifold structures.

In the context of fusing multisource data, several challenges persist. First, each modality employs independent encoders for feature extraction, leading to increased Parameters and computational complexity and introduces potential overfitting issues. Second, the information imbalance among multisource data undermines the effectiveness of feature interactions. For example, the richness of spatial–spectral information in HSI often outweighs the contribution from SAR or LiDAR data, leading to an imbalance in feature learning.

1) SIAM: When discussing the feature extraction method employed by SIAM, it is essential to first understand the core principle of this approach: determining the importance of information from each modality through GN operations and sharing weights between the two modalities to facilitate spatial information learning. This approach places particular emphasis on handling redundant and complementary information during the feature propagation process, aiming to enhance the effectiveness of multisource data fusion.

In the feature extraction process, the strategy involves comparing the scaling factors of GN layers with predefined thresholds to decide whether features should be retained or replaced. When the scaling factor falls below the threshold, the current feature is considered redundant and is replaced by the corresponding feature channel from another modality. This mechanism not only reduces redundant information but also ensures that the model captures more representative features.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:56:22 UTC from IEEE Xplore. Restrictions apply.

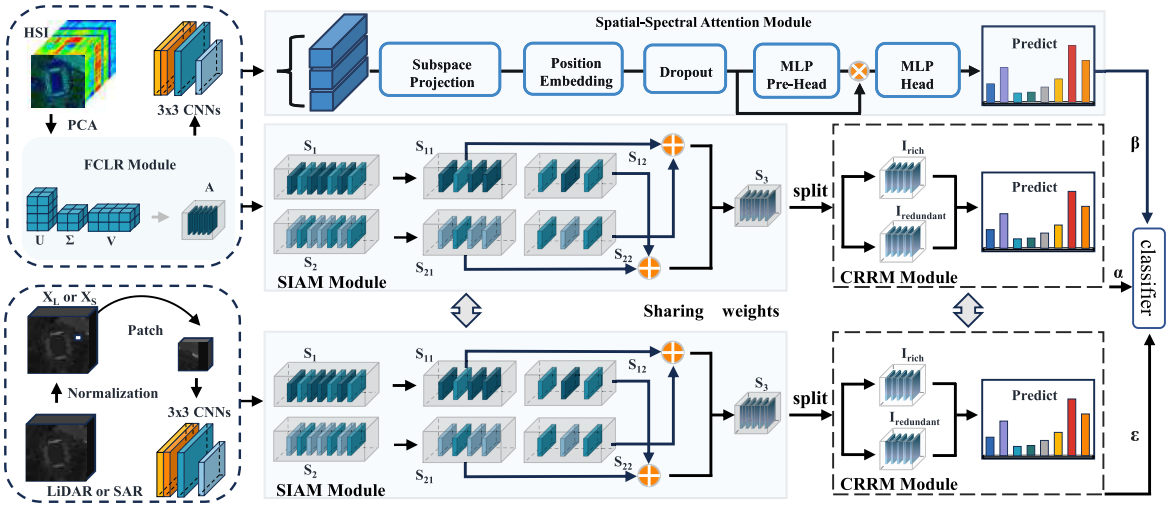


Fig. 2. Overview of the proposed ECNet framework. The framework includes (1) SIAM, (2) CRRM, (3) FCLR, (4) SSAM. During the decision fusion phase, the outputs of three classifiers are integrated using a network adaptive mechanism to achieve the ultimate classification decision. Here, the symbols α, β, and ε represent the weight coefficients acquired through adaptive learning.

Furthermore, SIAM can extract high-level semantic features from the input data shared in the feature space. This approach enables the network to map input data to a set of feature spaces, reducing the need for paired data and facilitating effective information correlation and fusion across multiple modalities. Since the weights are shared, data from the same category exhibit relatively consistent performance across various feature spaces, thus improving the model's generalization capability and robustness.

The specific representation is as follows:

$$\boldsymbol{\mu} = \frac{1}{N}\sum_{i=1}^{N} X_i \tag{1}$$

$$\boldsymbol{\sigma}^2 = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{x}_i - \boldsymbol{\mu})^2 \tag{2}$$

$$\mathbf{S} = \boldsymbol{\gamma} \odot \frac{\mathbf{x} - \boldsymbol{\mu}}{\sqrt{\boldsymbol{\sigma}^2 + \boldsymbol{\varepsilon}}} + \boldsymbol{\beta} \tag{3}$$

where $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ represent the mean and standard deviation of the input $\boldsymbol{x}$, respectively, and $\boldsymbol{\varepsilon}$ is a small constant introduced to prevent divisions by zero. We introduce learnable parameters $\boldsymbol{\gamma}$ and $\boldsymbol{\beta}$ for scaling and offset operations. Taking the partial derivative of the loss function L with respect to the parameter x provides gradient information. The specific formula is expressed as follows:

$$\begin{aligned}\frac{\partial L}{\partial \mathbf{x}} &= \frac{\partial L}{\partial \hat{\mathbf{x}}}\frac{\partial \hat{\mathbf{x}}}{\partial \mathbf{x}} + \frac{\partial L}{\partial \boldsymbol{\sigma}}\frac{\partial \boldsymbol{\sigma}}{\partial \mathbf{x}} + \frac{\partial L}{\partial \boldsymbol{\mu}}\frac{\partial \boldsymbol{\mu}}{\partial \mathbf{x}} \\ &= \frac{\partial L}{\partial \mathbf{y}}\frac{\partial \mathbf{y}}{\partial \hat{\mathbf{x}}} \cdot \left(\frac{\partial \hat{\mathbf{x}}}{\partial \boldsymbol{\mu}}\frac{\partial \boldsymbol{\mu}}{\partial \mathbf{x}} + \frac{\partial \hat{\mathbf{x}}}{\partial \boldsymbol{\sigma}}\frac{\partial \boldsymbol{\sigma}}{\partial \mathbf{x}} + \frac{\partial \hat{\mathbf{x}}}{\partial \mathbf{x}}\right) \\ &= \frac{\partial L}{\partial \mathbf{y}} \cdot \boldsymbol{\gamma} \cdot \left(\frac{\partial \hat{\mathbf{x}}}{\partial \boldsymbol{\mu}}\frac{\partial \boldsymbol{\mu}}{\partial \mathbf{x}} + \frac{\partial \hat{\mathbf{x}}}{\partial \boldsymbol{\sigma}}\frac{\partial \boldsymbol{\sigma}}{\partial \mathbf{x}} + \frac{\partial \hat{\mathbf{x}}}{\partial \mathbf{x}}\right).\end{aligned} \tag{4}$$

We use trainable parameters $\boldsymbol{\gamma}$ to assess the spatial pixel variances for each batch and channel. As the parameter $\boldsymbol{\gamma}$ tends toward zero, the gradient of the partial derivative $\partial L/\partial \mathbf{x}$ diminishes and it is necessary to adopt a fusion strategy that reduces redundancy. We employ normalized weights to denote the significance of various features. Through the application of the Sigmoid function, we map the weights to the (0, 1) interval, regulated by a threshold. If the weights surpass the threshold, they are set to 1; conversely, weights below the threshold are set to 0. The formula is articulated as follows:

$$\mathbf{W} = \begin{cases} 1, & \text{if Gate}\left(\dfrac{1}{1+\exp\left(-W_\gamma(GN(S))\right)}\right) \\ & > \text{threshold} \\ 0, & \text{if Gate}\left(\dfrac{1}{1+\exp\left(-W_\gamma(GN(S))\right)}\right) \\ & \leq \text{threshold}. \end{cases} \tag{5}$$

We employ cross-reconstruction operations to effectively combine and weight redundant and important information, thereby enhancing the information flow among different modalities. The entire operation is outlined as follows:

$$\begin{cases} \mathbf{S_1} = \mathbf{W_1} * \mathbf{S} \\ \mathbf{S_2} = \mathbf{W_2} * \mathbf{S} \\ \mathbf{S_{11}} + \mathbf{S_{12}} = \mathbf{S_1} \\ \mathbf{S_{21}} + \mathbf{S_{22}} = \mathbf{S_2} \\ [\mathbf{S_1}, \mathbf{S_2}] = \mathbf{S_3} \end{cases} \tag{6}$$

where $\mathbf{S_1}$ denotes a feature characterized by comprehensive and intricate information, whereas $\mathbf{S_2}$ signifies a feature with comparatively limited information.

2) CRRM: To effectively use feature information, we have introduced an innovative fusion mechanism that combines group convolution and point convolution, achieving a sparse fusion strategy. Group convolution aids in capturing global spatial patterns, while point convolution refines local spatial details. The integration of these two techniques enables the model to learn discriminative features across varying spatial scales. Specifically, we begin by partitioning the input feature into two distinct channels: αC and $(1-\alpha)C$, where α serves as the segmentation ratio factor. By distinguishing the importance of spatial information, we divide the data into two distinct feature representations, namely $\mathbf{I_{rich}}$ and $\mathbf{I_{redundant}}$.

We employed efficient convolution operations to enhance the representation of feature information in $\mathbf{I_{rich}}$. In this elevated transformation stage, our design emphasizes optimizing

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:56:22 UTC from IEEE Xplore. Restrictions apply.

information flow to enhance the effective capture of feature correlations

$$F_1 = \sum_{g=0}^{G-1} \text{conv}\left(\mathbf{I}, \mathbf{W_{group}}\right) \tag{7}$$

$$F_2 = M \odot \mathbf{I} \tag{8}$$

$$\mathbf{Y}_1 = F_1(\mathbf{I_{rich}}) + F_2(\mathbf{I_{rich}}) \tag{9}$$

where M denote the learnable weight matrices and G represents the number of group convolution.

In HSI data, various bands exhibit local correlations. Point convolution not only assists the model in capturing these local details but also enhances the spatial feature of complementary information. We feed $\mathbf{I_{redundant}}$ to generate feature maps with shallow hidden details, which are subsequently concatenated with the original $\mathbf{I_{redundant}}$ operation. This process allows the network to focus more on important features, thereby enhancing the representational capacity of the features

$$\mathbf{Y_2} = [F_2(\mathbf{I_{redundant}}), \mathbf{I_{redundant}}] \tag{10}$$

$$\mathbf{Y_3} = [\mathbf{Y_1}, \mathbf{Y_2}]. \tag{11}$$

Applying adaptive average pooling to $\mathbf{Y_3}$, normalize the result to obtain weights in the range of 0 to 1. Multiply these weights with the original $\mathbf{Y_3}$ to obtain the final result.

$$\mathbf{Out} = (\text{Softmax}(\text{Adapt}(\mathbf{Y_3})))\mathbf{Y_3}. \tag{12}$$

3) SSAM: The success of the spatial–spectral attention mechanism heavily relies on integrating and stacking multiple self-attention layers. With a focus on establishing long-range dependencies and extracting crucial features globally, the multihead self-attention (MHSA) mechanism [26] demonstrates significant applicability in both spatial and spectral domains. Renowned for its exceptional ability to capture internal feature correlations, this attention mechanism reduces dependence on external information. The employed parallel computing method significantly enriches the feature space, serving as a robust tool for extracting spatial and spectral features in HSI data. Importantly, this process incurs no additional computational time costs.

To be more specific, the intermediate features acquired through the convolution operation are initially rearranged as 1-D vectors. Subsequently, a learnable class embedding vector and positional encoding of each embedding pos are added to generate tokens in the patch-to-embedding process. The self-attention mechanism conducts scaled dot-product and Softmax operations on the input query Q and key K to create an attention map. This attention map is then subjected to dot-product and linear projection with the value V to derive the predicted class. Throughout this process, both high-level semantic information and long-tail abstract features are obtained simultaneously

$$\begin{aligned} Q &= \{Q_1, Q_2, \ldots, Q_i, \ldots, Q_s\} \\ K &= \{K_1, K_2, \ldots, K_i, \ldots, K_s\} \\ V &= \{V_1, V_2, \ldots, V_i, \ldots, V_s\} \end{aligned} \tag{13}$$

$$Z_i = \text{Attention}(Q_i, K_i, V_i) = \text{Softmax}\left(\frac{Q_i K_i^T}{\sqrt{d}}\right) V_i. \tag{14}$$

Multiple distinct SA layers can be combined to form a MHSA. Specifically, we initially obtain multiple attention representations and then concatenate them into a larger feature matrix, denoted as $[Z_1, Z_2, \ldots, Z_i, \ldots, Z_h]$. Finally, the feature dimensions are adjusted to match the input data through a linear transformation matrix W^O

$$\text{Output} = [Z_1, Z_2, \ldots, Z_i, \ldots, Z_h]W^O. \tag{15}$$

Layer normalization is applied prior to both the MHSA module and a multilayer perceptron layer within the encoder blocks. This strategic placement aims to improve the overall efficiency of the model. Additionally, the inclusion of residual connections acts as a valuable mechanism to mitigate the common issue of overfitting in DL methods. In summary, the computation within consecutive encoder blocks can be succinctly outlined as follows:

$$\tilde{z}_l = \text{MHSA}(\text{LayerNorm}(z_l)) + z_l \tag{16}$$

$$z_{l+1} = \text{MLP}\left(\text{LayerNorm}\left(\tilde{z}_l\right)\right) + \tilde{z}_l \tag{17}$$

where $\tilde{z}_l$ and z_l denote the output features of the MHSA module and MLP for block l, respectively.

4) FCLR: The forgetting curve, proposed as a theoretical model, characterizes the gradual attenuation of information over time. Originating from our understanding of human memory and information storage mechanisms, this concept reflects that individuals tend to forget or diminish their attention to past information as time progresses. From a cognitive perspective, the forgetting curve reveals the continuous evolution of information storage processes. In information processing, the challenge lies in accurately capturing the essential features of RS data. To address this issue, we chose a matrix approximation method based on the forgetting curve. Additionally, we implemented a truncated expansion of singular values to refine and compress the data representation. This strategy not only comprehensively considers the dynamic characteristics of information decay but also effectively enhances the modeling of the evolving properties of the data by simulating the trend of feature retention. The primary goal is to selectively retain pivotal information, providing a more refined input for subsequent analytical processes.

The formula for Ebbinghaus' forgetting curve is given by

$$R = e^{-\frac{t}{s}} \tag{18}$$

where R represents the retrievability of memory, s denotes memory stability, and t signifies time. Extending this concept to the realm of low-rank decomposition, we explore its inherent richness and broad applicability. In this context, the symbol t characterizes the width dimension of the data matrix X_H. Slices along different width dimensions reflect information pertaining to specific objects, textures, or edges. In the context of low-rank decomposition, a pivotal concept denoted as R is introduced to signify the proportion of redundant information relative to the original data in the decomposition. By retaining the most significant singular values and their corresponding singular vectors, a manifold representation of the original data is achieved. Additionally, the symbol s is introduced, representing the critical dimensions required for the upcoming feature extraction stage. In the context of multidimensional data, this representation underscores the importance of higher level feature extraction and data

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:56:22 UTC from IEEE Xplore. Restrictions apply.

analysis processes. Through a thorough analysis of network structure and data information, we adopt a strategy with fixed values for s and t. Drawing upon the Ebbinghaus forgetting curve, we have adaptively determined the range of values for R. In addition to offering flexibility in adjusting the degree of information preservation, the approach accommodates various tasks and data characteristics, underscoring the importance of efficiently extracting higher level features in multidimensional data processing. Through the implementation of this strategic approach, we attain a finer degree of control over the metric governing information preservation. Fig. 3 gives an illustration of how our proposed FCLR works.

Regarding the low-rank decomposition of a matrix, we can express it as follows:

$$\mathbf{B} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T. \tag{19}$$

$\mathbf{\Sigma}$ is a diagonal matrix defined as $\text{diag}(\sigma_1, \sigma_2, \ldots, \sigma_r)$, where r represents the rank of the matrix $\mathbf{\Sigma}$ and σ_i denotes the nonnegative singular values. The column vectors of $\boldsymbol{U}$ and $\boldsymbol{V}$ are unit orthogonal, representing the nonzero singular value's eigenvectors of $\boldsymbol{B}$. This expression captures the substantive characteristics of low-rank, indicating that through the decomposition of singular values, we can represent the original matrix $\boldsymbol{B}$ as the product of three matrices while maintaining the rank r of the matrix. The misclassification of pixels at class boundaries, leading to inadvertent smoothing of small targets, has been alleviated through the application of the FCLR, which employs LRMR techniques for the precise reconstruction of HSI data.

B. Global and Local Feature Fusion Architecture

Faced with the critical challenges in the field of information fusion, we propose a unique method that uses DL models to simultaneously merge multimodal RS data at both feature and decision levels. This strategy not only prioritizes the meticulous representation of low-level features but also emphasizes the precision of high-level decision-making, thereby achieving a more comprehensive and intelligent outcome.

We employ a weighted maximization fusion strategy to fully leverage the complementary information from each modality

$$F(\mathbf{X}_H, \mathbf{X}_C) = \max(w_h \cdot \mathbf{X}_H + w_c \cdot \mathbf{X}_C) \tag{20}$$

where X_H represents HSI data, and X_C represents supplementary RS data, such as LiDAR or SAR.

We use f to assist the network in making the final decision and the expression is as follows:

$$f(\mathbf{XW}) = \text{Softmax}(\mathbf{WX}) \tag{21}$$

$$\mathbf{O} = D\left[f_1(\mathbf{X}_H\mathbf{W}_H),\, f_2(\mathbf{X}_C\mathbf{W}_C),\, f_3(F(\mathbf{X}_H, \mathbf{X}_C)\mathbf{W}_F)\right]. \tag{22}$$

To emphasize global learning and obtain the final classification results through adaptive learning, the initial weights of f_1, f_2, and f_3 are set to be equal. This configuration aims to establish equitable initial conditions for the model, enabling autonomous learning of weight adjustments based on the diverse characteristics of the data. This optimization process enhances the global performance of decision-level fusion.

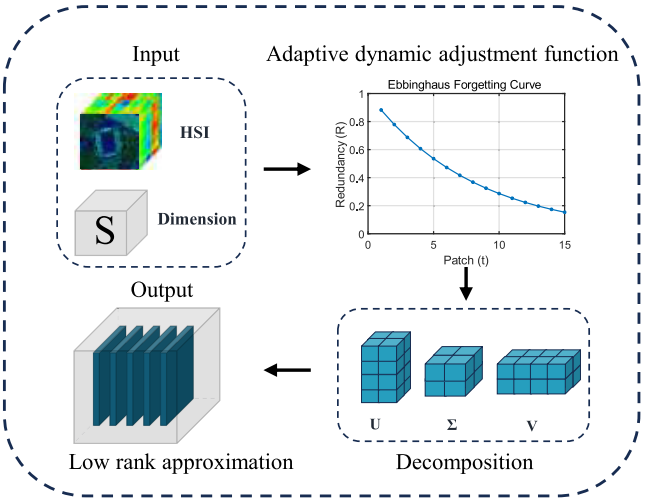


Fig. 3. Framework of FCLR.

C. Overall Loss Function

Based on the embedded semantic information in multisource RS data, we devised a similarity-driven loss function to optimize the feature decomposition of the model. After the feedforward process, we can obtain three outputs for each sample. Their loss values can be computed using the cross-entropy loss function. For instance, the loss value between the first output and the corresponding ground truth can be expressed as

$$L_1 = \frac{1}{N}\sum_i L_i = -\frac{1}{N}\sum_i \sum_{c=1}^{M} y_{ic}\log(p_{ic}) \tag{23}$$

M represents the number of types. The function y_{ic} is a symbolic function with two values: it takes 1 if the true class of sample i is equal to c, otherwise it takes 0. Here, p_{ic} represents the predicted probability that observation i belongs to class c.

The L_1 loss function is employed for spatial–spectral reconstruction in supervised HSI data analysis, whereas the L_2 loss function is responsible for feature learning in heterogeneous RS data. Simultaneously, the L_3 oversees the entire process of feature fusion in RS data. Incorporating the L_1 and L_2 norms as regularization terms in the L_3 model aims to effectively mitigate the risk of overfitting in the model.

The final output expression is as follows:

$$L = \lambda_1 L_1 + \lambda_2 L_2 + L_3 \tag{24}$$

in this context, λ_1 and λ_2 denote the weighting parameters associated with L_1 and L_2, respectively. The formulation of this loss function takes into account the significance of each network's loss, culminating in final optimization through the meticulous adjustment of weighting parameters.

IV. Experiments and Analysis

A. Data Description

In this section, we evaluate the performance of the proposed network using four distinct HSI modalities along with their associated multimodal data sources (LiDAR, SAR). The experimental datasets include the Houston2013 Dataset, the Trento Dataset, the MUUFL Gulfport scene Dataset, and the Augsburg Dataset. For detailed information, refer to Table I.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:56:22 UTC from IEEE Xplore. Restrictions apply.

TABLE I
DESCRIPTION OF THE FOUR DATASETS

Dataset	**Houston2013**		**Trento**		**MUUFL**		**Augsburg**	
Sensor Type	HSI	LiDAR	HSI	LiDAR	HSI	LiDAR	HSI	SAR
Image Size	349×1905	349×1905	600×166	600×166	325×220	325×220	332 ×485	332×485
Number of Bands	144	1	63	1	64	2	180	2
Wavelength Range	0.38-1.05 μm	/	0.42-0.99 μm	/	375-1050 nm	/	0.4-2.5 μm	/
Spatial Resolution	2.5 m	2.5 m	1 m	1 m	0.54×1.0 m	0.60×0.78m	30 m	30 m

B. Evaluation Metrics and Parameter Setting

1) Evaluation Metrics: Three widely used metrics, overall accuracy (OA), average accuracy (AA), accuracy for each category (CA), and Kappa coefficient (κ) were applied to quantitatively assess the performance of land use and land cover (LULC) classification methods. The detailed calculation formulas for Evaluation Metrics can be found in [9].

2) Parameter Setting: We implemented the proposed network model on the PyTorch platform, conducting training and testing on an NVIDIA A100 GPU with 80 GB of memory to meet the demands of large-scale DL models. Specifically, we used the Adam optimizer for parameter updates, set the batch size to 64, and established a learning rate of e^{-3}. For model training, we performed 200 epochs on each dataset. The choice of epoch number was based on a comprehensive evaluation of model performance, ensuring stable and satisfactory results across diverse datasets. To ensure a fair comparison of batch FLOPs and Parameters across different algorithms, we standardized the batch size to 1 in our experiments.

C. Comparisons With Previous Methods

A set of state-of-the-art DL algorithms has been selected for comparison with ECNet. CCR-Net [7] is a typical dual-branch model designed to achieve the fusion of multimodal data. coupled_CNN [27] employs a weight-sharing mechanism to achieve deep interaction among multimodal data, enhancing classification performance. ExViT [28] introduces an attention mechanism to effectively integrate information from both LiDAR and HSI by weighting the features. LSAF [29] proposes a linear self-attention fusion model, incorporating an adaptive decision fusion module to aggregate classification results. HCT [30] uses a hierarchical architecture for hierarchical fusion and classification of LiDAR and HSI data. Additionally, thorough consideration has been given to the latest mainstream multimodal fusion network architectures, including SSEFT [31], MHST [32], and DSHF [33]. The selected methodology encompasses various DL paradigms, such as dual-branch structures, weight sharing, attention mechanisms, hierarchical frameworks, and decision-level fusion strategies, among others. These are employed to meet the modeling requirements for the complex interdependencies in heterogeneous RS data. We assess the performance of these methods through a series of quantitative metrics, including single-category classification accuracy, OA, AA, and κ. Additionally, we conduct visual evaluations to further appraise the classification effectiveness of each method.

The accuracy performance of our proposed model, along with these comparison method, on the Houston2013, Trento, MUUFL, and Augsburg datasets can be found in Tables II–V. The best results are highlighted in bold. Figs. 4–7 display the visual effects on different datasets, respectively. The visual metrics indicate that the proposed framework performs optimally across multiple datasets and generally outperforms other methods in terms of accuracy. Using the Houston2013 dataset as an example, Table II demonstrates the superior performance of the coupled_CNN network with a weight-sharing mechanism over purely convolutional feature extraction networks CCR, highlighting the importance of targeted feature extraction. Furthermore, the positive impact of attention mechanisms in feature-level fusion is validated, as seen in the ExViT network. The application of a suitable decision-level fusion strategy, as exemplified by LSAF, effectively enhances classification results. We seamlessly integrate the aforementioned advantages into an end-to-end framework, where we not only facilitate comprehensive multimodal information interaction but also employ attention mechanisms atop feature-level fusion. Additionally, we incorporate decision-level fusion strategies, effectively enhancing the classification performance. Comparatively, our method achieves the best performance in OA, AA, and κ on the Houston2013 dataset, attaining the highest classification accuracy in six out of the 15 categories. It is evident that for the Houston2013 dataset, our method outperforms the suboptimal approach with an OA improvement of 2.26%, AA improvement of 1.78%, and κ improvement of 2.46%. The proposed framework achieves an outstanding OA of 96.63%, showcasing competitiveness in joint classification tasks involving HSI and LiDAR data. Particularly noteworthy are the significant performance enhancements observed in residential and commercial areas.

Regarding the Trento dataset, we have achieved outstanding performance, with a κ reaching an impressive level of 98.97%, signifying exceptional accuracy. In the MUUFL dataset, particularly for the ClouthPanels category with limited training samples, the DSHF failed to correctly identify this category, while our ECNet achieved a remarkable accuracy of 93.36%.

Additional experiments were conducted on the Augsburg dataset using HSI and SAR data for comparison. Quantitative and visual classification results are presented in Table V and Fig. 7, respectively. The performance gap between feature extraction networks relying solely on convolution, such as CCR-Net, and other advanced classifiers is more pronounced compared to the previous Houston2013 dataset. In terms of OA, CCR-Net achieves only 86.47%, while the latest

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:56:22 UTC from IEEE Xplore. Restrictions apply.

TABLE II
OA, AA, AND KAPPA COEFFICIENT ON THE HOUSTON2013 DATASET BY CONSIDERING HSI AND LIDAR DATA

	Total (2832/12197)	**coupled_CNN**	**CCR-Net**	**LSAF**	**HCT**	**MHST**	**SSEFT**	**ExViT**	**DSHF**	**ECNet**	**ECNet**
	Sensor Type	(H+L)	(H+L)	(H+L)	(H+L)	(H+L)	(H+L)	(H+L)	(H+L)	(H)	(H+L)
1	Healthy grass (198/1053)	82.81	88.41	83.10	**97.34**	92.18	83.38	82.91	87.56	85.19	96.30
2	Stressed grass (190/1064)	99.81	**99.91**	83.08	96.62	96.80	97.09	98.68	99.34	98.03	**99.91**
3	Synthetic grass(192/505)	97.23	**100.00**	**100.00**	84.75	97.82	97.23	99.60	98.81	88.12	99.01
4	Tree(188/1056)	**100.00**	99.15	89.49	96.78	99.81	98.86	99.15	99.81	97.35	98.86
5	Soil (186/1056)	**100.00**	99.34	**100.00**	**100.00**	**100.00**	99.72	99.91	99.91	98.39	99.53
6	Water (182/143)	**100.00**	95.80	**100.00**	96.50	96.50	91.61	99.30	93.71	95.80	**100.00**
7	Residential (196/1072)	92.35	94.96	92.72	82.09	95.52	91.32	96.08	83.96	86.75	**97.20**
8	Commercial (191/1053)	94.30	91.26	92.69	95.54	96.39	90.03	90.03	80.63	85.47	**97.34**
9	Road (193/1059)	92.73	92.63	**97.07**	90.84	87.72	80.36	86.12	64.68	85.17	93.20
10	Highway (191/1036)	85.14	79.63	68.44	58.88	82.24	55.31	72.97	**96.72**	86.39	91.60
11	Railway (181/1054)	**98.86**	94.21	89.94	97.53	98.01	84.82	88.99	78.75	85.48	98.39
12	Park lot 1 (192/1041)	93.28	88.38	**96.25**	90.11	85.49	77.71	90.39	86.36	86.36	92.22
13	Park lot 2 (184/285)	89.47	77.19	89.12	**97.19**	92.98	61.75	90.18	88.77	86.67	88.42
14	Tennis court (181/247)	**100.00**	96.76	**100.00**	**100.00**	94.33	94.74	99.60	98.79	91.09	99.60
15	Running track (187/473)	98.94	99.79	**100.00**	**100.00**	97.04	92.18	95.14	**100.00**	95.35	**100.00**
	OA(%)	94.37	93.15	90.51	91.15	93.81	86.33	91.40	89.01	89.69	**96.63**
	AA(%)	94.99	93.16	92.13	92.28	94.19	86.41	92.60	90.52	90.11	**96.77**
	$\kappa(\times 100)$	93.88	92.56	89.69	90.40	93.28	85.18	90.66	88.06	88.81	**96.34**

TABLE III
OA, AA, AND KAPPA COEFFICIENT ON THE TRENTO DATASET BY CONSIDERING HSI AND LIDAR DATA

	Total (819/29395)	**coupled_CNN**	**CCR-Net**	**LSAF**	**HCT**	**MHST**	**SSEFT**	**ExViT**	**DSHF**	**ECNet**	**ECNet**
	Sensor Type	(H+L)	(H+L)	(H+L)	(H+L)	(H+L)	(H+L)	(H+L)	(H+L)	(H)	(H+L)
1	Apples (129/3905)	99.31	**100.00**	99.41	98.67	**100.00**	98.46	98.82	**100.00**	99.05	99.97
2	Buildings(125/2778)	96.69	98.67	87.51	**99.92**	99.78	97.95	98.42	98.92	96.22	98.16
3	Ground (105/374)	91.18	90.37	91.71	**99.73**	94.92	89.30	79.68	79.95	71.93	89.84
4	Woods (154/8969)	99.74	99.91	99.14	99.96	99.80	**100.00**	99.99	99.58	97.68	99.82
5	Vineyard (184/10317)	**100.00**	94.76	99.89	67.39	99.83	96.83	99.84	97.69	99.79	99.89
6	Roads (122/3052)	91.78	85.39	95.22	86.57	84.76	89.18	92.37	**96.56**	81.75	93.91
	OA(%)	98.55	96.37	97.84	86.96	98.21	97.23	98.58	98.35	96.48	**98.97**
	AA(%)	96.45	94.85	95.48	92.04	96.52	95.29	94.85	95.45	91.07	**96.93**
	$\kappa(\times 100)$	98.06	95.16	97.10	83.23	97.60	96.30	98.10	97.79	95.28	**98.62**

Transformer architecture, SSEFT achieves an accuracy of only 84.42%, underscoring the increased challenge posed by this dataset. In all competitive methods, our network exhibits superior performance in terms of OA, AA, and κ, with visual results notably aligning closely with the GT.

D. Hyperparameter Sensitivity Analysis

Model complexity stands as another crucial metric, and in this section, we conduct a quantitative analysis on the complexity of the model. Specifically, we employ FLOPs to analyze the time complexity of the model and use the number of Parameters to gauge its space complexity. As illustrated in Fig. 1, the diagram depicts the OA and computational complexity across various network architectures. The radius of each circle corresponds to the magnitude of the parameter quantity. The proposed ECNet exhibits significantly fewer Parameters and FLOPs than the majority of networks. Compared to the second-best-performing coupled_CNN, ECNet exhibits slightly higher FLOPs, yet it boasts a reduced

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:56:22 UTC from IEEE Xplore. Restrictions apply.

TABLE IV

OA, AA, AND KAPPA COEFFICIENT ON THE MUUFL GULFPORT SCENE BY CONSIDERING HSI AND LIDAR DATA

	Total (2683/51004)	**coupled_CNN**	**CCR-Net**	**LSAF**	**HCT**	**MHST**	**SSEFT**	**ExViT**	**DSHF**	**ECNet**	**ECNet**
	Sensor Type	(H+L)	(H+L)	(H+L)	(H+L)	(H+L)	(H+L)	(H+L)	(H+L)	(H)	(H+L)
1	Trees (1162/22084)	97.67	97.44	97.00	97.03	**98.86**	97.37	98.58	95.06	96.63	98.59
2	Grass-Pure (214/4056)	88.09	77.93	83.36	90.29	79.83	77.76	87.70	**92.38**	81.04	90.68
3	Grass-Groundsurface (344/6538)	88.99	82.90	90.79	90.07	79.63	83.13	**90.96**	72.90	89.60	90.62
4	Dirt-And-Sand (91/1735)	**95.22**	88.13	94.18	94.18	93.95	82.54	90.61	94.87	93.89	94.64
5	Road-Materials (334/6353)	97.32	96.49	95.88	93.86	94.84	95.17	94.73	90.41	93.88	**97.34**
6	Water (23/443)	**99.10**	94.36	94.58	95.71	92.78	92.10	93.68	1.81	90.74	97.29
7	Buildings'Shadow (112/2121)	85.38	83.36	87.36	87.09	87.13	78.97	90.05	**97.97**	80.10	83.59
8	Buildings (312/5928)	98.08	97.93	98.14	96.61	97.12	96.78	97.76	92.70	97.40	**98.65**
9	Sidewalk (69/1316)	51.60	52.89	**75.91**	46.35	68.77	63.22	68.54	56.08	55.55	61.55
10	Yellow-Curb (9/174)	10.34	2.30	13.22	18.97	**28.16**	26.44	23.56	0.00	14.94	18.97
11	ClothPanels (13/256)	81.64	85.16	84.77	75.39	**93.75**	91.02	80.86	0.00	89.06	93.36
	OA(%)	93.65	91.50	93.70	92.95	92.43	91.17	94.37	88.65	92.03	**94.77**
	AA(%)	81.22	78.08	83.20	80.50	83.17	80.41	83.37	63.11	80.26	**84.12**
	$\kappa(\times 100)$	91.59	88.74	91.68	90.69	89.92	88.32	92.54	85.15	89.46	**93.06**

TABLE V

OA, AA, AND KAPPA COEFFICIENT ON THE AUGSBURG DATASET BY CONSIDERING HSI AND SAR DATA

	Total (761/77533)	**coupled_CNN**	**CCR-Net**	**LSAF**	**HCT**	**MHST**	**SSEFT**	**ExViT**	**DSHF**	**ECNet**	**ECNet**
	Sensor Type	(H+S)	(H+S)	(H+S)	(H+S)	(H+S)	(H+S)	(H+S)	(H+S)	(H)	(H+S)
1	Forest (146/13361)	92.96	92.06	97.21	94.23	**98.92**	91.83	93.51	97.06	96.82	97.42
2	Commercial Area (7/1638)	1.77	6.78	0.31	4.82	1.40	9.34	**15.20**	8.91	8.30	8.73
3	Residential Area (264/30065)	97.04	97.13	**98.97**	98.54	94.55	90.19	97.31	96.88	96.64	95.56
4	Industrail Area (21/3830)	**76.66**	62.09	30.31	43.79	64.33	52.22	64.26	14.83	73.32	75.07
5	Low Plants (248/26609)	96.10	84.20	95.97	95.33	86.04	88.29	87.57	**98.01**	93.03	96.70
6	Allotment (52/523)	50.10	46.27	49.14	**67.88**	52.01	55.64	52.77	5.54	54.49	54.49
7	Water (23/1507)	30.66	26.94	11.94	**52.09**	34.37	8.63	27.01	47.51	25.48	43.07
	OA(%)	91.39	86.47	90.13	90.90	87.47	84.42	88.28	89.81	90.75	**92.13**
	AA(%)	63.61	59.35	54.84	65.24	61.66	56.59	62.52	52.68	64.01	**67.29**
	$\kappa(\times 100)$	87.56	80.63	85.45	86.83	82.17	77.74	83.28	84.93	86.71	**88.70**

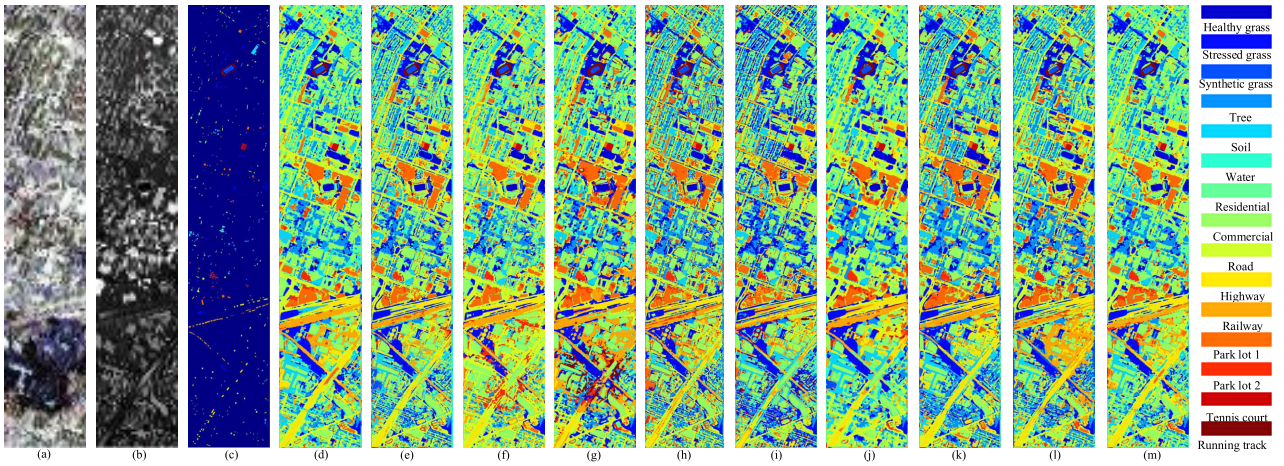


Fig. 4. Visualization of false-color HSI and LiDAR images using different comparison methods based on the Houston2013 dataset. (a) HSIMAP (H). (b) LiDARMAP (L). (c) GT (H + L). (d) Coupled_CNN (H + L). (e) CCR-Net (H + L). (f) LSAF (H + L). (g) HCT (H + L). (h) MHST (H + L). (i) SSEFT (H + L). (j) ExViT (H + L). (k) DSHF (H + L). (l) ECNet (H). (m) ECNet (H + L).

parameter count. The performance enhancements of ECNet effectively compensate for this aspect and the substantial increase in classification accuracy rationalizes the parameter trade-off.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:56:22 UTC from IEEE Xplore. Restrictions apply.

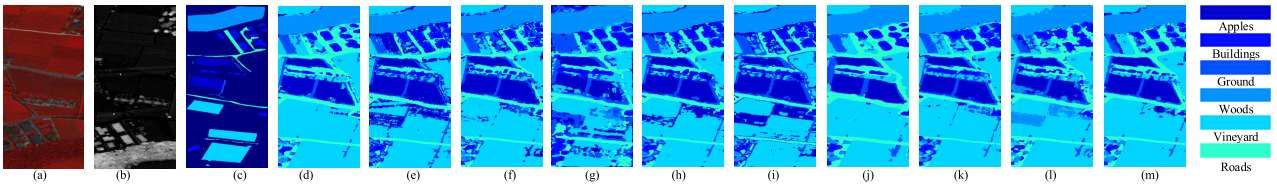


Fig. 5. Visualization of false-color HSI and LiDAR images using different comparison methods based on the Trento dataset. (a) HSIMAP (H). (b) LiDARMAP (L). (c) GT (H + L). (d) Coupled_CNN (H + L). (e) CCR-Net (H + L). (f) LSAF (H + L). (g) HCT (H + L). (h) MHST (H + L). (i) SSEFT (H + L). (j) ExViT (H + L). (k) DSHF (H + L). (l) ECNet (H). (m) ECNet (H + L).

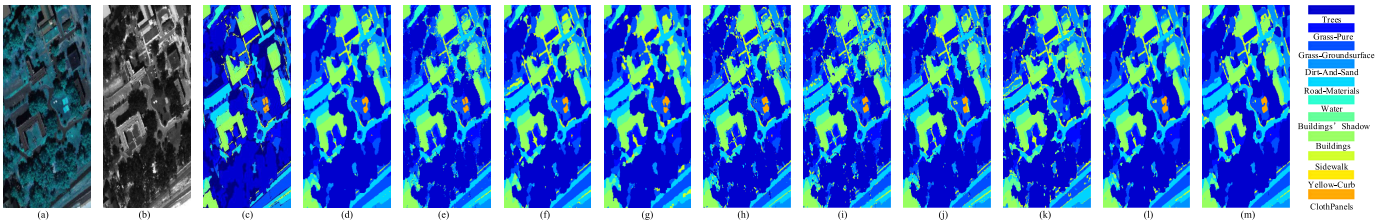


Fig. 6. Visualization of false-color HSI and LiDAR images using different comparison methods based on the MUUFL Gulfport scene dataset. (a) HSIMAP (H). (b) LiDARMAP (L). (c) GT (H + L). (d) Coupled_CNN (H + L). (e) CCR-Net (H + L). (f) LSAF (H + L). (g) HCT (H + L). (h) MHST (H + L). (i) SSEFT (H + L). (j) ExViT (H + L). (k) DSHF (H + L). (l) ECNet (H). (m) ECNet (H + L).

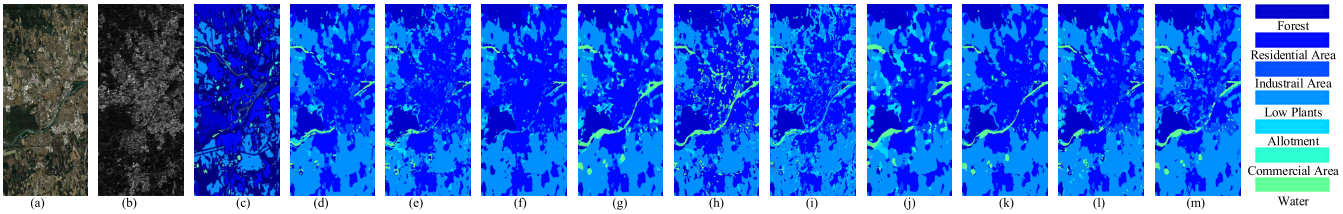


Fig. 7. Visualization of false-color HSI and SAR images using different comparison methods based on the Augsburg dataset. (a) HSIMAP (H). (b) LiDARMAP (L). (c) GT (H + L). (d) Coupled_CNN (H + L). (e) CCR-Net (H + L). (f) LSAF (H + L). (g) HCT (H + L). (h) MHST (H + L). (i) SSEFT (H + L). (j) ExViT (H + L). (k) DSHF (H + L). (l) ECNet (H). (m) ECNet (H + L).

TABLE VI
ABLATION EXPERIMENTS ON FOUR MULTISOURCE RS DATASETS WITH DIFFERENT MODULES OF OA(%) AA(%) AND KAPPA

FCLR	SIAM	CRRM	SSAM	Houston2013			Trento		
				OA(%)	AA(%)	$\kappa(\times 100)$	OA(%)	AA(%)	$\kappa(\times 100)$
✓	✓	✓	✓	**96.63**	**96.77**	**96.34**	**98.97**	**96.93**	**98.62**
✗	✓	✓	✓	92.24(↓4.39)	92.23((↓4.54)	91.57(↓4.77)	97.68(↓1.29)	93.88(↓3.05)	96.90(↓1.72)
✓	✗	✓	✓	94.61(↓2.02)	95.17(↓1.60)	94.14(↓2.20)	98.12(↓0.85)	94.04(↓2.89)	97.47(↓1.15)
✓	✓	✗	✓	95.27(↓1.36)	95.66(↓1.11)	94.86(↓1.48)	98.16(↓0.81)	95.37(↓1.56)	97.54(↓1.08)
✓	✓	✓	✗	94.55(↓2.08)	94.37(↓2.40)	94.08(↓2.26)	98.11(↓0.86)	94.29(↓2.64)	97.46(↓1.16)
FCLR	**SIAM**	**CRRM**	**SSAM**	**MUUFL**			**Augsburg**		
				OA(%)	AA(%)	$\kappa(\times 100)$	OA(%)	AA(%)	$\kappa(\times 100)$
✓	✓	✓	✓	**94.77**	**84.12**	**93.06**	**92.13**	**67.29**	**88.70**
✗	✓	✓	✓	94.05(↓0.72)	82.22(↓1.90)	92.10(↓0.96)	90.25(↓1.88)	60.21(↓7.08)	85.98(↓2.72)
✓	✗	✓	✓	93.70(↓1.07)	80.63(↓3.49)	91.66(↓1.40)	91.55(↓0.58)	64.99(↓2.30)	87.91(↓0.79)
✓	✓	✗	✓	94.61(↓0.16)	82.41(↓1.71)	92.86(↓0.20)	91.85(↓0.28)	63.59(↓3.70)	88.19(↓0.51)
✓	✓	✓	✗	93.27(↓1.50)	78.96(↓5.16)	91.04(↓2.02)	90.70(↓1.43)	63.98(↓3.31)	86.69(↓2.01)

E. Ablation Study

1) Comparison of the Performance of Different Modules: To explore performance enhancements and validate the rationale behind our proposed ECNet. As detailed in Section III, ECNet comprises modules for FCLR, SIAM, CRRM, and SSAM. Subsequently, we conducted further ablation experiments on these different feature modules to assess their individual impact. The quantitative analysis in Table VI suggests that the effective utilization of different modules can lead to varying levels of performance enhancement.

2) Validate the Accuracy of the Forgetting Curve in the FCLR Module: The parameter N in FCLR denotes the number of retained principal singular values, directly influencing both the model's performance and computational efficiency.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:56:22 UTC from IEEE Xplore. Restrictions apply.

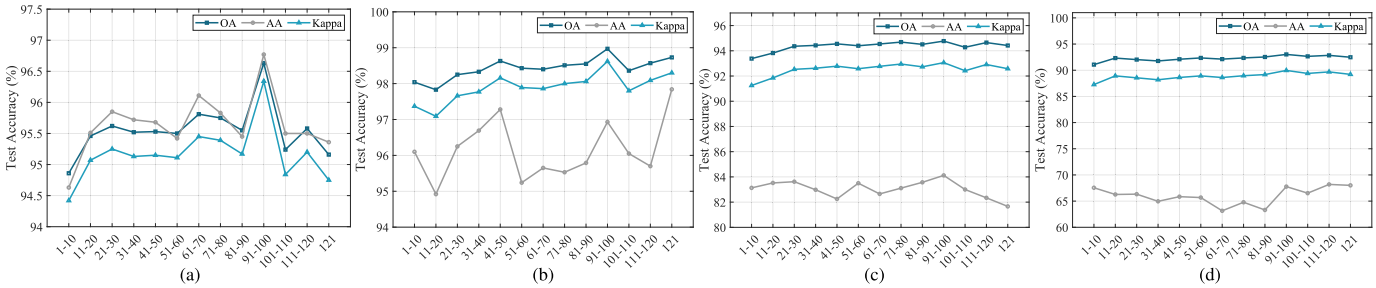


Fig. 8. Visualization of accuracy of ECNet using different N of FCLR With Patch = 11. (a) Houston. (b) Trento. (c) MUUFL. (d) Augsburg.

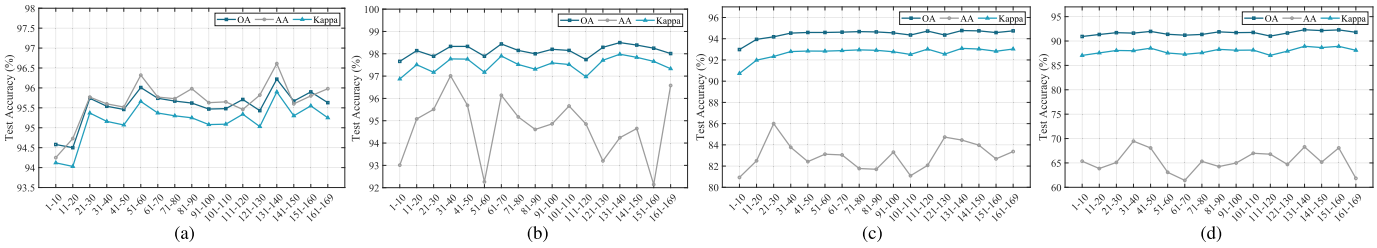


Fig. 9. Visualization of accuracy of ECNet using different N of FCLR With Patch = 13. (a) Houston. (b) Trento. (c) MUUFL. (d) Augsburg.

Optimal selection of N is imperative to strike a balance between model complexity and generalization capability. Larger values of N may introduce challenges associated with overfitting, while smaller values may result in information loss, albeit reducing computational and storage burdens. Hence, selecting an appropriate N involves balancing the retention of sufficient information with computational costs. Through an in-depth investigation into the relationship between patch size and OA, we identified the optimal range for selecting the parameter N within the context of FCLR. Illustrated in Figs. 8 and 9, our investigation was guided by the Ebbinghaus forgetting curve theory, which addresses information loss.

We delved into the efficacy of FCLR in reducing redundancy in the data by examining various intensities of input patches and their corresponding performance accuracies, ultimately deriving a self-learned N value. Analysis of the charts reveals that when the patch size is set to 11, the optimal classification performance corresponds to an N value in the range of 90–100. Similarly, for a patch size of 13, optimal performance is achieved with an N value ranging from 130 to 140. These findings corroborate the self-learned N values, validating the effectiveness of redundancy reduction based on the Ebbinghaus forgetting curve and underscoring the broader significance of spectral dimensionality. This systematic inquiry not only aids in elucidating the rationale behind N value selection but also furnishes specific parameter ranges crucial for optimizing model performance across distinct tasks and data contexts. By ensuring the model captures critical information while mitigating computational burdens, our approach significantly enhances classification effectiveness.

V. Conclusion

This article propose a novel network for the collaborative classification of multisource RS data. In the feature extraction stage, to alleviate the DL issue of misclassification of pixels located at class boundaries leading to smoothing of small targets, we employ a manifold feature preservation mechanism based on the Ebbinghaus forgetting curve achieving meticulous spectral reconstruction for each cubic block within HSI. In addition, we efficiently combine the SIAM and CRRM paradigms to create a more compact representation of features across different modalities, accommodating varying data complexities. During the fusion stage, a combined fusion strategy of feature-level fusion and decision-level fusion is applied. Not only has our proposed framework demonstrated significant achievements in handling HSI and LiDAR data through extensive data validation, but it has also been successfully extended to more challenging scenarios involving HSI and SAR data, achieving remarkable results. In future work, we plan to explore more efficient and robust network designs to achieve faster and higher performance multisource RS data fusion. Additionally, we aim to integrate our ECNet into other tasks, such as super-resolution target detection and weakly supervised semantic segmentation. Effective coupling of complementary information from multisource observations and reduction of redundant features are crucial for enhancing the effectiveness of these applications.

References

[1] J. Li et al., "Deep learning in multimodal remote sensing data fusion: A comprehensive review," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 112, Aug. 2022, Art. no. 102926.

[2] J. Liang, D. Hu, Y. Wang, R. He, and J. Feng, "Source data-absent unsupervised domain adaptation through hypothesis transfer and labeling transfer," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 8602–8617, Aug. 2021.

[3] J. Hu, D. Hong, Y. Wang, and X. X. Zhu, "A comparative review of manifold learning techniques for hyperspectral and polarimetric SAR image fusion," *Remote Sens.*, vol. 11, no. 6, p. 681, Mar. 2019.

[4] D. Hong, L. Gao, R. Hang, B. Zhang, and J. Chanussot, "Deep encoder–decoder networks for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[5] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 937–949, Feb. 2018.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:56:22 UTC from IEEE Xplore. Restrictions apply.

[6] M. Zhang, W. Li, R. Tao, H. Li, and Q. Du, "Information fusion for classification of hyperspectral and LiDAR data using IP-CNN," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2021.

[7] X. Wu, D. Hong, and J. Chanussot, "Convolutional neural networks for multimodal remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5517010.

[8] S. Mei, R. Jiang, M. Ma, and C. Song, "Rotation-invariant feature learning via convolutional neural network with cyclic polar coordinates convolutional layer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5600713.

[9] D. Hong et al., "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, Aug. 2020.

[10] S. Khan, M. Naseer, M. Hayat, S. W. Zamir, F. S. Khan, and M. Shah, "Transformers in vision: A survey," *ACM Comput. Surv.*, vol. 54, no. 10s, pp. 1–41, Jan. 2022.

[11] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2021.

[12] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Jan. 2022, Art. no. 5522214.

[13] S. Mei, C. Song, M. Ma, and F. Xu, "Hyperspectral image classification using group-aware hierarchical transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5539014.

[14] S. C. Kulkarni and P. P. Rege, "Pixel level fusion techniques for SAR and optical images: A review," *Inf. Fusion*, vol. 59, pp. 13–29, Jul. 2020.

[15] P. Benedetti, D. Ienco, R. Gaetano, K. Ose, R. G. Pensa, and S. Dupuy, "M^3Fusion: A deep learning architecture for multiscale multimodal multitemporal satellite data fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 12, pp. 4939–4949, Dec. 2018.

[16] B. Xi et al., "DGSSC: A deep generative spectral–spatial classifier for imbalanced hyperspectral imagery," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 4, pp. 1535–1548, Apr. 2023.

[17] Y. Cai, Z. Zhang, Z. Cai, X. Liu, X. Jiang, and Q. Yan, "Graph convolutional subspace clustering: A robust subspace clustering framework for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4191–4202, May 2021.

[18] Y. Bu et al., "Hyperspectral and multispectral image fusion via graph Laplacian-guided coupled tensor decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 648–662, May 2020.

[19] F. Luo, Z. Zou, J. Liu, and Z. Lin, "Dimensionality reduction and classification of hyperspectral image via multistructure unified discriminative embedding," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5517916.

[20] Y. Lu, C. Yuan, X. Li, Z. Lai, D. Zhang, and L. Shen, "Structurally incoherent low-rank 2DLPP for image classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 29, no. 6, pp. 1701–1714, Jun. 2019.

[21] J. Peng, L. Li, and Y. Y. Tang, "Maximum likelihood estimation-based joint sparse representation for the classification of hyperspectral remote sensing images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 6, pp. 1790–1802, Jun. 2019.

[22] J. Peng, W. Sun, and Q. Du, "Self-paced joint sparse representation for the classification of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1183–1194, Feb. 2019.

[23] D. Li, W. Xie, Y. Li, and L. Fang, "FedFusion: Manifold-driven federated learning for multi-satellite and multi-modality fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5500813.

[24] A. Yuan, J. Yu, X. Zuo, K. Gao, and M. You, "Low rank and semi-nonnegative tensor factorization for hyperspectral imagery denoising," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 1153–1156.

[25] Y. Lu, S. Mei, F. Xu, M. Ma, and X. Wang, "Separable deep graph convolutional network integrated with CNN and prototype learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516216.

[26] C. H. van Berkel, "Multi-core for mobile phones," in *Proc. Design, Autom. Test Eur. Conf. Exhib.*, Apr. 2009, pp. 1260–1265.

[27] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020.

[28] J. Yao, B. Zhang, C. Li, D. Hong, and J. Chanussot, "Extended vision transformer (ExViT) for land use and land cover classification: A multimodal deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514415.

[29] M. Feng, F. Gao, J. Fang, and J. Dong, "Hyperspectral and LiDAR data classification based on linear self-attention," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jun. 2021, pp. 2401–2404.

[30] G. Zhao, Q. Ye, L. Sun, Z. Wu, C. Pan, and B. Jeon, "Joint classification of hyperspectral and LiDAR data using a hierarchical CNN and transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500716.

[31] Y. Feng, J. Zhu, R. Song, and X. Wang, "S2EFT: Spectral–spatial-Elevation fusion transformer for hyperspectral image and LiDAR classification," *Knowl.-Based Syst.*, vol. 283, Jan. 2024, Art. no. 111190.

[32] K. Ni, D. Wang, Z. Zheng, and P. Wang, "MHST: Multiscale head selection transformer for hyperspectral and LiDAR classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 5470–5483, 2024.

[33] Y. Feng, L. Song, L. Wang, and X. Wang, "DSHFNet: Dynamic scale hierarchical fusion network based on multiattention for hyperspectral image and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522514.



Weiying Xie (Senior Member, IEEE) received the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in 2017.

She is currently a Professor with the State Key Laboratory of Integrated Services Networks, Xidian University. She has published more than 50 articles in refereed journals and proceedings, including IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTION ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON CYBERNETICS, and Conference on IEEE/CVF Computer Vision and Pattern Recognition Conference (CVPR) and Association for the Advancement of Artificial Intelligence (AAAI). Her research interests include neural networks, machine learning, hyperspectral image processing, and high-performance computing.



Yibing Lu received the B.E. degree in software engineering from Xidian University, Xi'an, China, in 2023, where she is currently pursuing the M.S. degree with the Image Coding and Processing Center, State Key Laboratory of Integrated Service Network.

Her research interests include multimodal image processing, remote sensing classification, and target detection.



Daixun Li received the B.E. degree in telecommunications engineering from Xidian University, Xi'an, China, in 2022, where he is currently pursuing the M.S. degree with the Image Coding and Processing Center, State Key Laboratory of Integrated Service Network.

His research interests include multimodal image processing, remote sensing classification, and federated learning.



Yunsong Li (Member, IEEE) received the M.S. degree in telecommunication and information systems and the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 1999 and 2002, respectively.

He joins the school of Telecommunications Engineering, Xidian University, in 1999, where he is currently a Professor. He is the Director of the Image Coding And Processing Center, State Key Laboratory of Integrated Service Networks. His research interests include image and video processing and high-performance computing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:56:22 UTC from IEEE Xplore. Restrictions apply.